# Neural Networks

## Introduction

Slides are partially adapted from notes by Vincent Cheung, Kevin Cannons and William Kaiser

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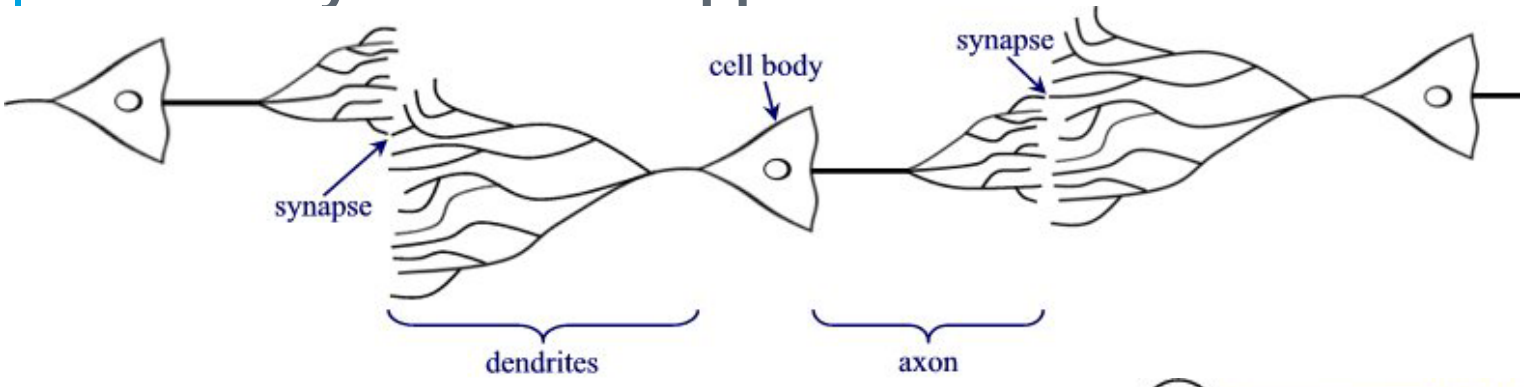
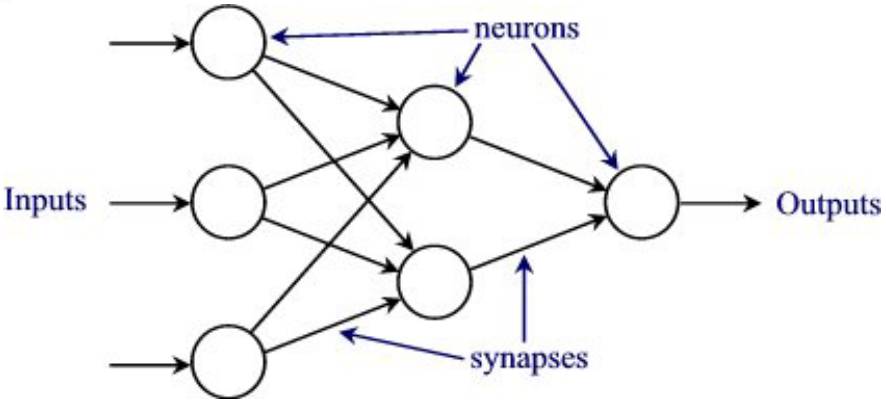
# What are Artificial Neural Networks?

- An extremely simplified model of the brain
- Composed of many “neurons” that co-operate to perform the desired function
- Essentially a function approximator



# What are they used for?

## Classification

- Pattern recognition, feature extraction, image matching

## Noise reduction

- Recognize patterns in the inputs and produce noiseless outputs

## Prediction

- Extrapolation based on historical data

# Why use Neural Networks?

## Ability to learn

- Can figure out how to perform their function on their own
- Determine their function based only upon sample inputs

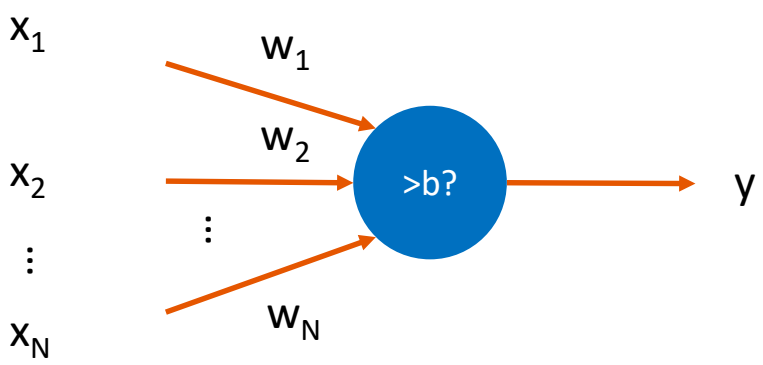
## Ability to generalize

- Produce reasonable outputs for inputs it has not been taught how to deal with

# Perceptrons

- First neural network with the ability to learn
- Made up of a single neuron
- The first perceptron (from which modern neural networks evolved) was invented by a psychologist, Frank Rosenblatt, in 1958.
- This marked a major achievement in the history of artificial intelligence.

# How does a Perceptron work?



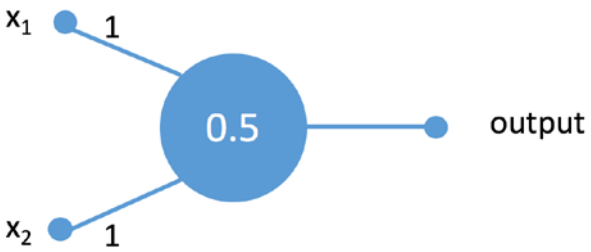

$y = 1$ if $x_1w_1 + x_2w_2 + \ldots + x_Nw_N > b$
$y = 0$ if $x_1w_1 + x_2w_2 + \ldots + x_Nw_N \leq b$

- **$x_1$, $x_2$, …, $x_N$ are inputs, y is output.**
- **$w_1$, $w_2$, …, $w_N$ are weights that represent the strength of each connection to the neuron.**
- **b is the bias, a threshold to determine whether a neuron is activated by the inputs.**

# Function: OR

Perception can be designed to perform the function of a OR gate:

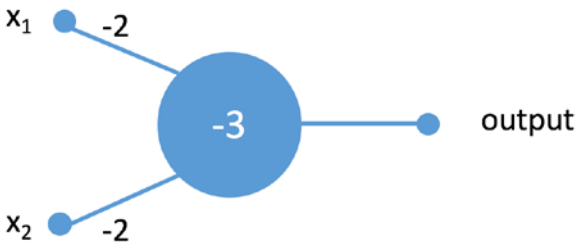
| $x_1$ | $x_2$ | output |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |



# Function: NAND

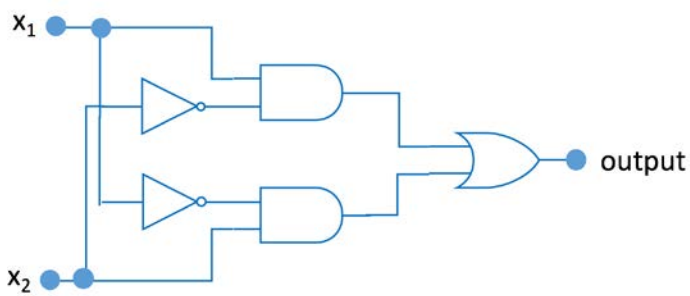
## Another example of a NAND gate:

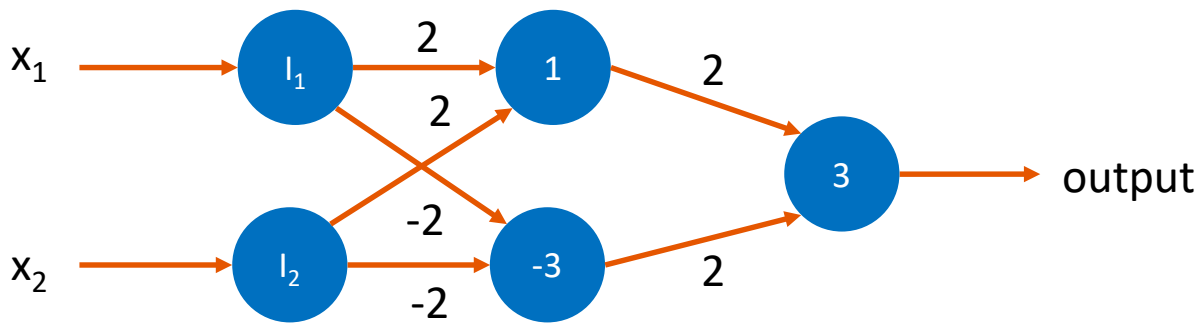
| $x_1$ | $x_2$ | output |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |



# What about XOR?

## More complex functions need more complex networks.

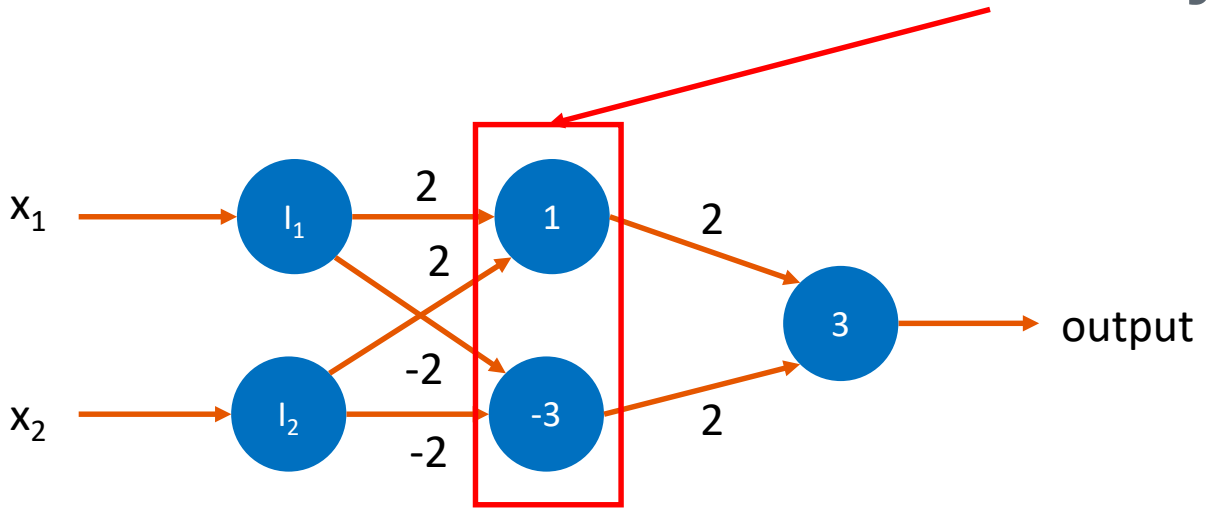



| $x_1$ | $x_2$ | output |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

# Hidden Layer

- The XOR Gate can be implemented with a new neural network structure that includes additional neurons that are between the network inputs and network output.
- The new structure includes a “Hidden Layer”.



# Question

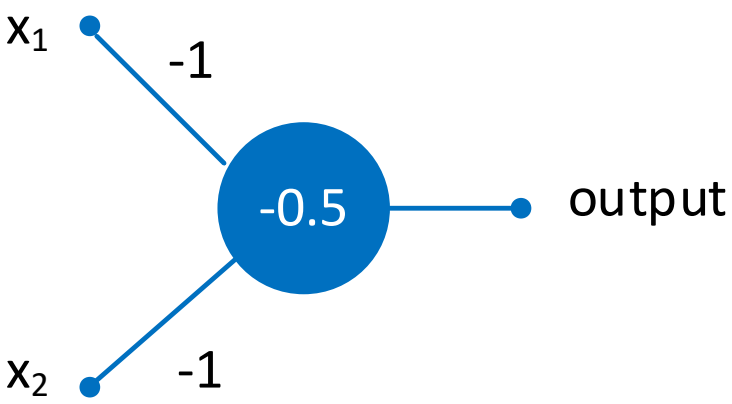
## What function does the perceptron implement?

a) AND

b) OR

c) NAND

d) NOR



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